

A Surjective Isometry Satisfying the Kernel–Range Condition but Having No Semigroup Embedding

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Source question

Eisner’s paper *Embedding operators into strongly continuous semigroups*, arXiv:0808.2419, proves that an embeddable bounded operator T must have both $\dim \ker T$ and $\operatorname{codim} \operatorname{rg} T$ equal to either zero or infinity. Question 3.4 asks for which operator classes this condition is sufficient. Later, Remark 5.4 explicitly requests a non-trivial operator satisfying this necessary condition which is not embeddable.

Take $x_n \in \ker T\left(\frac{1}{2^n}\right)$ with $\|x_n\| = 1$ for every $n \in \mathbb{N}$. Then we have $\{x_n\}_{n=1}^\infty \subset \ker T$ which is finite-dimensional. Therefore there exists a subsequence $\{x_{n_k}\}_{k=1}^\infty$ converging to some x_0 with $\|x_0\| = 1$.

Since $\ker T\left(\frac{1}{2^{n+1}}\right) \subset \ker T\left(\frac{1}{2^n}\right)$, $n \in \mathbb{N}$, we have $T\left(\frac{1}{2^n}\right)x_0 = 0$ for every $n \in \mathbb{N}$ which contradicts the strong continuity of $T(\cdot)$.

Assume now that $0 < \operatorname{codim}(\operatorname{rg} T) = \dim(\ker T') < \infty$ holds. If T is embedded into a C_0 -semigroup $T(\cdot)$, then T' is embedded into the adjoint semigroup $T'(\cdot)$ on X' which is weak* continuous. The same arguments as above lead to a contradiction to the weak* continuity. $\square$

A direct corollary is the following.

Corollary 3.2. *Non-bijective Fredholm-operators are not embeddable.*

Remark 3.3. As we will see in Section 4, all fore possibilities given in Theorem 3.1 (dimension 0 or ∞ of the kernel and codimension 0 or ∞ of the range) can appear for operators with the embedding property. The examples will be unitary operators, the left and right shifts on $l^2(Y)$ for an infinite-dimensional Hilbert space Y and their direct sum, respectively.

In the following we ask the converse question.

Question 3.4. For which (classes of) operators the necessary condition given in Theorem 3.1 is also sufficient for embeddability?

Result

The necessary condition is not sufficient, even in the much narrower class of invertible surjective isometries on complex Banach spaces. An elementary counterexample is multiplication by the coordinate function on $C(\mathbb{T})$. In fact, this operator has no n -th root for any $n \geq 2$, so its failure of embeddability is purely algebraic and does not depend on continuity estimates for a candidate semigroup.

Proof

Let $X = C(\mathbb{T})$, with the supremum norm, and define

$$(Vf)(z) = zf(z) \quad (z \in \mathbb{T}).$$

The inverse is multiplication by $\bar{z}$. Hence V is an invertible surjective isometry. In particular,

$$\dim \ker V = 0, \quad \text{codim rg } V = 0,$$

so it satisfies Eisner's necessary condition in its strongest possible form.

Lemma 1. *If $S \in \mathcal{L}(C(\mathbb{T}))$ and $S^n = V$ for some integer $n \geq 2$, then there is a function $h \in C(\mathbb{T})$ such that $h^n(z) = z$ for every $z \in \mathbb{T}$.*

Proof. Because $S^n = V$, the operators S and V commute. Since V is invertible, S also commutes with every integer power V^k , $k \in \mathbb{Z}$. Thus S commutes with multiplication M_p by every trigonometric polynomial p . Trigonometric polynomials are uniformly dense in $C(\mathbb{T})$, and $\|M_p - M_f\| = \|p - f\|_\infty$; consequently S commutes with M_f for every $f \in C(\mathbb{T})$.

Put $h = S\mathbf{1}$. For arbitrary $f \in C(\mathbb{T})$,

$$Sf = S(M_f\mathbf{1}) = M_f(S\mathbf{1}) = fh.$$

Thus $S = M_h$. Applying $S^n = V$ to $\mathbf{1}$ gives $h^n(z) = z$ on $\mathbb{T}$. □

Theorem 1. *The surjective isometry V has no bounded n -th root for any integer $n \geq 2$, and therefore cannot be embedded into a C_0 -semigroup.*

Proof. If $h^n(z) = z$, then h never vanishes and $|h(z)| = 1$ on $\mathbb{T}$. Let $d \in \mathbb{Z}$ be the winding number of $h : \mathbb{T} \rightarrow \mathbb{T}$. The winding number of h^n is nd , whereas the coordinate function $z \mapsto z$ has winding number 1. The equality $h^n(z) = z$ would therefore imply $nd = 1$, impossible for $n \geq 2$. Lemma 1 now excludes every bounded n -th root of V .

If $(T(t))_{t \geq 0}$ were a C_0 -semigroup with $T(1) = V$, then $T(1/n)^n = T(1) = V$ for every $n \geq 2$, contradicting the preceding conclusion. □

Corollary 1. *Eisner's kernel-range condition is not sufficient for embeddability on general complex Banach spaces, even after restricting to invertible surjective isometries.*

This also separates sharply the Banach-space situation from Theorem 4.4 of the source paper: every unitary operator on a Hilbert space is embeddable, whereas a surjective isometry on a Banach space need not even possess a square root.

Relation to the explicit request in Remark 5.4

such that $\sigma(T_0) = \{0\}$ and $\sigma(T_j) = \{\lambda_j\}$ for $\lambda_j \neq 0$. Every T_j can be embedded by spectral calculus. Since $r(T_0) = 0$, the operator T_0 is sectorial by the Neumann series representation for the resolvent and hence can be embedded as well by Theorem 2.3. So the original operator T can be embedded into a C_0 -semigroup. $\square$

As an example we see that the Volterra operator $(Vf)(\tau) = \int_0^\tau f(s)ds$ on $X = C[0, 1]$ has the embedding property.

Remark 5.4. It is not clear whether a compact operator with dense (or infinite-codimensional) range and infinite-dimensional kernel can always be embedded into a C_0 -semigroup. (Note that finite-dimensional kernel and cofinite-dimensional range is excluded by Theorem 3.1.) Furthermore, it is desirable to construct a non-trivial operator (in the sense of Theorem 3.1) which cannot be embedded into a C_0 -semigroup.

The operator V is “non-trivial in the sense of Theorem 3.1” exactly as requested: neither a finite nonzero kernel nor a finite nonzero range defect explains the failure. The obstruction instead lies in the topology of the maximal multiplication algebra generated by V .

Scope and upgrade audit

This packet gives a complete counterexample to sufficiency for arbitrary operators and answers the source’s request for a non-trivial example. It does not settle the separate compact-operator question in Remark 5.4, nor does it classify all operator classes for which the condition is sufficient.

The strongest route was checked at each vulnerable point: a hypothetical semigroup supplies bounded roots; an arbitrary root must commute with V ; commutation with V and V^{-1} extends by trigonometric-polynomial density to every multiplier; and the commutant calculation then forces the root to be a multiplier. The final winding-number obstruction excludes all orders $n \geq 2$, not only square roots. Human review should check the short commutant lemma and confirm novelty; the mathematical claim otherwise uses only norm-density of trigonometric polynomials and $\deg(h^n) = n \deg(h)$.

Novelty and duplicate checks

The run’s cheap indexes were searched for arXiv 0808.2419, the source title, operator embeddability, strongly continuous semigroups, roots, and multiplication operators. No exact packet or attempt was found. Bounded searches of primary literature found the source paper and later work on special classes (including real/positive operators and Toeplitz operators), but no arXiv paper identifying this $C(T)$ counterexample as an answer to Question 3.4 or Remark 5.4. This is therefore recorded as a candidate new elementary counterexample, subject to human novelty review.

References

- [1] T. Eisner, *Embedding operators into strongly continuous semigroups*, Archiv der Mathematik 92 (2009), 451–460; arXiv:0808.2419.